Package 'fracreg'

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Type Package

Title Fractional Response Models

License GPL-3

Description An R package for the estimation and specification analysis of fractional response models. Provides routines for fitting univariate one-part, two-part, and double-inflated three-part fractional models. Further incorporates estimators for panel data settings and addresses unobserved heterogeneity and endogeneity via correlated random effects and control function approaches. Calculates analytical partial effects across all model types and includes generalised goodness-of-functional-form (GGOFF) and RESET hypothesis tests.

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Depends R (>= 3.5.0)

Contents

`fracreg-package` *Fractional Response Models*

Description

Provides comprehensive tools for the estimation and specification analysis of fractional response models. It supports univariate one-part, hurdle two-part, and double-inflated three-part models. The package also incorporates estimators for panel data settings (CRE, GMM, QML) and addresses unobserved heterogeneity and endogeneity via correlated random effects and control function approaches. It supports various link functions, calculates average and conditional partial effects analytically across all model types, and provides robust specification tests such as RESET, P test, and GGOFF tests.

Details

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License:	GPL-3

Acknowledgements

This package builds upon, consolidates, and modernises the fractional regression frameworks originally implemented in the `frm`, `frmhet`, and `frmpd` R packages developed by Joaquim J.S. Ramalho. As those original packages have been deprecated and removed from the active CRAN repository, `fracreg` serves as an actively maintained successor, ensuring these econometric tools remain available to the R community.

Author(s)

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References

Ramalho, J. J. S. *frm: Fractional Regression Models*. R package. Formerly available on CRAN, currently archived.

Ramalho, J. J. S. *frmhet: Fractional Regression Models under Heterogeneity*. R package. Formerly available on CRAN, currently archived.

Ramalho, J. J. S. *frmpd: Fractional Regression Models for Panel Data*. R package. Formerly available on CRAN, currently archived.

Papke, L. E. and Wooldridge, J. M. (1996), "Econometric methods for fractional response variables with an application to 401(k) plan participation rates", *Journal of Applied Econometrics*, 11(6), 619-632.

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional response models", *Journal of Economic Surveys*, 25(1), 19-68.

Ramalho, E. A., Ramalho, J. J. S., and Murteira, J. M. R. (2014), "A two-part fractional response model for the spatial distribution of vineyards in Portugal", *Journal of Applied Econometrics*, 29(4), 607-630.

Fang, K., & Ma, S. (2013), "Three-part model for fractional response variables with application to Chinese household health insurance coverage", *Journal of Applied Statistics*, 40(5), 925-940.

`fracreg` *Fitting Fractional Response Models*

Description

`fracreg` is used to fit fractional response models, which are appropriate for responses that are proportions, percentages, or fractions restricted to the [0, 1] interval. It supports standard one-part models, two-part hurdle models for modelling boundary values at 0 or 1, and three-part models for double inflation at both 0 and 1.

Usage

```
fracreg(y, x, x2 = x, linkbin, linkfrac, type = "1P", inflation = 0,
        intercept = TRUE, table = TRUE, variance = TRUE, var.type = "default",
        var.eim = TRUE, var.cluster, dfc = FALSE, ...)
```

Arguments

`y`	a numeric vector containing the values of the response variable.
`x`	a numeric matrix, with column names, containing the values of the covariates.
`x2`	a numeric matrix, with column names, containing the values of the covariates in the fractional component of two-part models if option `type = "2P"` is defined. Defaults to `x`.
`linkbin`	a description of the link function to use in the binary component of a two-part fractional response model, or a vector of two link functions for the two binary components of a three-part model (e.g. `c("logit", "probit")`). Available options: `logit`, `probit`, `cauchit`, `loglog`, `cloglog`.
`linkfrac`	a description of the link function to use in standard fractional response models or in the fractional component of a two-part fractional response model. Available options: `logit`, `probit`, `cauchit`, `loglog`, `cloglog`.
`type`	a description of the model to estimate: a standard one-part model (`1P`, the default), a two-part model (`2P`), the binary component of a two-part model (`2Pbin`), the fractional component of a two-part model (`2Pfrac`), or a three-part model (`3P`) for double boundary inflation.
`inflation`	a numeric value indicating which of the extreme values of `0` (the default) or `1` is the relevant boundary value for defining two-part fractional response models.
`intercept`	a logical value indicating whether the model should include a constant term or not.
`table`	a logical value indicating whether a summary table with the regression results should be printed.

`variance` a logical value indicating whether the variance of the estimated parameters should be calculated. Defaults to `TRUE` whenever `table = TRUE`.

`var.type` a description of the type of variance of the estimated parameters to be calculated. Options are `standard` (recommended for models estimated by maximum likelihood, such as the binary component of two-part models), `robust` (recommended for models estimated by quasi-maximum likelihood, such as standard fractional response models or the fractional component of a two-part fractional response model), `cluster` (recommended in the case of panel data) and `default` (implements the `standard` or `robust` versions as appropriate).

`var.eim` a logical value indicating whether the expected information matrix should be used in the calculation of the variance. When false, the observation information matrix will be used. Defaults to `TRUE`.

`var.cluster` a numeric vector containing the values of the variable that specifies to which cluster each observation belongs.

`dfc` a logical value indicating whether a degrees of freedom correction should be applied to the covariance matrix. Defaults to `FALSE`.

`...` Arguments to pass to glm.

Details

`fracreg` estimates one-part, two-part hurdle, and three-part double-inflated fractional response models; see Ramalho, Ramalho and Murteira (2011) and Fang and Ma (2013) for details on those models.

One-Part Fractional Response Models (`type = "1P"`): The standard one-part model assumes that the conditional expectation of the fractional response $y_i \in [0, 1]$ is given by:

$$E(y_i|x_i) = G(x_i\beta)$$

where $G(\cdot)$ is a known non-linear link function mapping the linear predictor to the unit interval (e.g., logit, probit). The parameters β are estimated by maximising the Bernoulli-based quasi-log-likelihood function:

$$\ln L_i(\beta) = y_i \ln[G(x_i\beta)] + (1 - y_i) \ln[1 - G(x_i\beta)]$$

This estimator requires only the correct specification of the conditional mean to yield consistent parameter estimates (Papke and Wooldridge, 1996).

Two-Part Hurdle Models (`type = "2P"`): When the data exhibits a boundary mass (e.g., at $y_i = 0$), the two-part hurdle model handles the boundary values separately from the interior fractional values. Let y_i^* be a binary indicator such that $y_i^* = 1$ if $y_i > 0$ and $y_i^* = 0$ otherwise. The probability of observing a boundary value is modelled as:

$$P(y_i = 0|x_{1i}) = 1 - F(x_{1i}\gamma_1)$$

$$P(y_i > 0|x_{1i}) = F(x_{1i}\gamma_1)$$

where $F(\cdot)$ is a binary link function. Conditional on observing an interior fractional value, the response is modelled as:

$$E(y_i|x_{2i}, y_i > 0) = G(x_{2i}\beta_2)$$

The unconditional mean of the response is therefore:

$$E(y_i|x_i) = F(x_{1i}\gamma_1) \times G(x_{2i}\beta_2)$$

Three-Part Double Inflated Models (`type = "3P"`**):** For data containing boundary mass at both 0 and 1, the three-part model estimates two separate binary mechanisms for each boundary and a fractional component for the interior values $(0, 1)$, extending the two-part logic to double inflation (Fang and Ma, 2013).

`fracreg` uses the standard glm command to perform the estimations. Therefore, `fracreg` is essentially a convenience command, allowing estimation of several alternative fractional response models using the same command. In addition, `fracreg` provides an R-squared measure for all models (calculated as the square of the correlation coefficient between the actual and fitted values of the dependent variable), calculates the fitted values of the dependent variable in two-part models and stores the information needed to implement some very useful commands for fractional response models: fracreg.reset (RESET test), fracreg.ptest (P test), fracreg.ggoff (GGOFF tests) and fracreg.pe (partial effects).

Value

When `type = "1P"` or `"2Pfrac"`, `fracreg` returns a list with the following elements:

`class`	"fracreg".
`formula`	the model formula.
`type`	the name of the estimated model.
`link`	the name of the specified link.
`method`	estimation method. Currently, "QML" (quasi-maximum likelihood) for fractional components or models and"ML" (maximum likelihood) for the binary component of two-part models.
`p`	a named vector of coefficients.
`yhat`	the fitted mean values.
`xbhat`	the fitted mean values of the linear predictor.
`converged`	logical. Was the algorithm judged to have converged?
`x.names`	a vector containing the names of the covariates.

If `variance = TRUE` or `table = TRUE`, the previous list also contains the following elements:

`p.var`	a named covariance matrix.
`var.type`	covariance matrix type.
`var.eim`	logical. Was the expected information matrix used in the computation of the covariance matrix?
`dfc`	logical. Was a degrees of freedom correction used for the computation of the covariance matrix?

If `var.type = "cluster"`, the list also contains the following element:

`var.cluster`	the variable that specifies to which cluster each observation belongs.

When `type = "2Pbin"`, `fracreg` returns a similar list with the following additional element:

`LL`	the value of the log-likelihood.

When `type = "2P"`, `fracreg` returns the previous lists, indexed by the prefixes `resBIN` and `resFRAC`, and the following additional elements:

`class`	"fracreg".
`type`	"2P".

ybase	a numeric vector containing the values of the response variable.
x2base	a numeric matrix containing the values of the covariates.
yhat2P	the overall fitted mean values.
converged	logical. Were the algorithms judged to have converged in both parts of the model?

When type = "3P", fracreg returns the previous lists, indexed by the prefixes resBIN0, resBIN1, and resFRAC, and the following additional elements:

class	"fracreg".
type	"3P".
ybase	a numeric vector containing the values of the response variable.
x2base	a numeric matrix containing the values of the covariates.
yhat3P	the overall fitted mean values.
converged	logical. Were the algorithms judged to have converged in all parts of the model?

Author(s)

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References

Papke, L. E. and Wooldridge, J. M. (1996), "Econometric methods for fractional response variables with an application to 401(k) plan participation rates", *Journal of Applied Econometrics*, 11(6), 619-632.

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional response models", *Journal of Economic Surveys*, 25(1), 19-68.

Fang, K., & Ma, S. (2013), "Three-part model for fractional response variables with application to Chinese household health insurance coverage", *Journal of Applied Statistics*, 40(5), 925-940.

See Also

fracreg.reset and fracreg.ggoff, for specification tests.
fracreg.ptest, for non-nested hypothesis tests.
fracreg.pe, for computing partial effects.
fracreghet, for fitting cross-sectional fractional response models with unobserved heterogeneity.
fracregpd, for fitting panel data fractional response models.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age,
           totemp = fracreg_k401k$totemp, sole = fracreg_k401k$sole)

# 1P Model
fracreg(y, X, type="1P", linkfrac="logit")

# 2P Model (modelling mass at 1)
fracreg(y, X, type="2P", inflation=1, linkbin="logit", linkfrac="logit")
```

```
# 3P Model (inject artificial 0s for demonstration)
y_3p <- y; y_3p[1:50] <- 0
fracreg(y_3p, X, type="3P", linkbin=c("logit","logit"), linkfrac="logit")

### Simulated Examples

set.seed(123)
N <- 1000
x1 <- rnorm(N)
x2 <- runif(N)

# Generating a fractional dependent variable with inflation at 0 and 1
XB <- -0.5 + 0.8 * x1 + 1.2 * x2 + rnorm(N)
y_latent <- exp(XB) / (1 + exp(XB))

y <- y_latent
# Inflate at boundaries
y[y_latent < 0.2] <- 0
y[y_latent > 0.8] <- 1

X <- cbind(x1 = x1, x2 = x2)

# fracreg estimation of a logit fractional response model
fracreg(y, X, type="1P", linkfrac="logit")

# fracreg estimation of the binary logit component of the two-part fractional
# regression model with y=0 as the relevant boundary value
fracreg(y, X, type="2Pbin", inflation=0, linkbin="logit")

# fracreg estimation of the fractional component of the two-part fractional
# regression model with y=0 as the relevant boundary value and using a
# probit link function
fracreg(y, X, type="2Pfrac", inflation=0, linkfrac="probit")

# fracreg estimation of both components of a two-part fractional response model
# with y=0 as the relevant boundary value and using a cloglog binary link
# function and a logit fractional link function
fracreg(y, X, type="2P", inflation=0, linkbin="cloglog", linkfrac="logit")

# Three-part double-inflated model (y has both 0s and 1s)
fracreg(y, X, type="3P", linkbin=c("logit","probit"), linkfrac="logit")
```

`fracreg.ggoff` *GGOFF Tests for Fractional Response Models*

Description

`fracreg.ggoff` is used to perform Generalised Goodness-Of-Functional-Form (GGOFF) tests to check the adequacy of the functional form and link specification of fractional response models.

Usage

```
fracreg.ggoff(object, version = "LM", table = TRUE, ...)
```

Arguments

object	an object containing the results of an fracreg command.
version	a vector containing the test versions to use. Available options: Wald, LM (the default) and, only for the binary component of two-part models, LR. More than one option may be chosen.
table	a logical value indicating whether a summary table with the test results should be printed.
...	Arguments to pass to glm, which is used to estimate the model under the alternative hypothesis when version is a vector containing "Wald" or "LR".

Details

fracreg.ggoff applies the GGOFF, GOFF1 and GOOFF2 test statistics to fractional response models estimated via fracreg. fracreg.ggoff may be used to test the link specification of: (i) one-part fractional response models; (ii) the binary component of two-part fractional response models; and (iii) the fractional component of two-part fractional response models.

GGOFF Test Framework: The Generalised Goodness-of-Functional Form (GGOFF) test evaluates the adequacy of the link function $G(\cdot)$. It is based on augmenting the baseline model with specific directions of departure. The auxiliary testing equation takes the form:

$$E(y|x) = G\left(x\beta + \gamma_1 \frac{g'(x\hat{\beta})}{g(x\hat{\beta})} + \gamma_2 x\hat{\beta}\right)$$

where $g(\cdot)$ and $g'(\cdot)$ are the first and second derivatives of $G(\cdot)$ evaluated at the linear predictor $x\hat{\beta}$. The test checks $H_0 : \gamma_1 = 0, \gamma_2 = 0$. GOFF1 and GOFF2 are variants testing individual components.

When the Wald version is implemented, it is taken into account the option that was chosen for computing standard errors in the model under evaluation. For the LM version, a robust version is computed in cases (i) and (iii) and a conventional version in case (ii). See Ramalho, Ramalho and Murteira (2014) for details on the application of the GGOFF, GOFF1 and GOOFF2 tests in the fractional response framework.

Value

fracreg.ggoff returns a named vector with the test results.

Author(s)

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References

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2014), "A generalized goodness-of-functional form test for binary and fractional response models", *Manchester School*, 82(4), 488-507.

Pregibon, D. (1980), "Goodness of Link Tests for Generalized Linear Models", *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 29(1), 15-24.

See Also

fracreg, for fitting fractional response models.
fracreg.reset, for asymptotically equivalent specification tests.
fracreg.ptest, for non-nested hypothesis tests.
fracreg.pe, for computing partial effects.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age,
           totemp = fracreg_k401k$totemp, sole = fracreg_k401k$sole)

m <- fracreg(y, X, type="1P", linkfrac="logit")
fracreg.ggoff(m)

### Simulated Examples

N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#Testing the logit specification of a standard fractional response model
#using LM and Wald versions of the GGOFF test, based on 1 or 2 fitted powers of
#the linear predictor
res <- fracreg(y,X,linkfrac="logit",table=FALSE)
fracreg.ggoff(res,c("Wald","LM"))

#Testing the probit specification of the binary component of a two-part fractional
#regression model using a LR-based GGOFF test
res <- fracreg(y,X,linkbin="probit",type="2Pbin",inf=1,table=FALSE)
fracreg.ggoff(res,"LR")
```

---

fracreg.pe — *Fractional Response Models - Partial Effects*

Description

`fracreg.pe` is used to compute average and/or conditional partial effects in fractional response models.

Usage

```
fracreg.pe(object, APE = TRUE, CPE = FALSE, at = NULL, which.x = NULL,
           variance = TRUE, table = TRUE)
```

Arguments

`object`	an object containing the results of an `fracreg` command.
`APE`	a logical value indicating whether average partial effects are to be computed.
`CPE`	a logical value indicating whether conditional partial effects are to be computed.

`at`	a numeric vector containing the covariates' values at which the conditional partial effects are to be computed or the strings `"mean"` (the default) or `"median"`, in which cases the covariates are evaluated at their mean or median values (or mode, in case of dummy variables), respectively.
`which.x`	a vector containing the names of the covariates to which the partial effects are to be computed.
`variance`	a logical value indicating whether the variance of the estimated partial effects should be calculated. Defaults to `TRUE` whenever `table = TRUE`.
`table`	a logical value indicating whether a summary table with the results should be printed.

Details

`fracreg.pe` calculates partial effects for fractional response models estimated via `fracreg`. `fracreg.pe` may be used to compute average or conditional partial effects for: (i) one-part fractional response models; (ii) the binary components of two-part and three-part fractional response models; (iii) the fractional components of two-part and three-part fractional response models; and (iv) two-part and three-part fractional response models overall.

Partial Effects for Continuous Variables: For a continuous covariate x_k, the partial effect on the conditional mean $E(y|x) = G(x\beta)$ is the first derivative with respect to x_k:

$$PE_k(x) = \frac{\partial E(y|x)}{\partial x_k} = g(x\beta)\beta_k$$

where $g(\cdot)$ is the probability density function corresponding to the link function $G(\cdot)$.

Partial Effects for Discrete Variables: For a discrete or dummy covariate x_k, the partial effect is calculated as the discrete difference in the expected value when x_k changes from 0 to 1, holding all other variables x_{-k} constant:

$$PE_k(x) = G(x_{-k}\beta_{-k} + \beta_k) - G(x_{-k}\beta_{-k})$$

Average vs. Conditional Partial Effects: - **Average Partial Effects (APE):** Evaluated for each observation i in the sample and then averaged:

$$APE_k = \frac{1}{N}\sum_{i=1}^{N} PE_k(x_i)$$

- **Conditional Partial Effects (CPE):** Evaluated at a specific vector of covariate values x^* (e.g., the sample mean or median):

$$CPE_k = PE_k(x^*)$$

For calculating standard errors, it is taken into account the option that was previously chosen for estimating the model. See Ramalho, Ramalho and Murteira (2011) and Fang and Ma (2013) for details on the computation of partial effects in the fractional response framework.

Value

`fracreg.pe` returns a list with the following element:

`PE.p`	a named vector of partial effects.

If `variance = TRUE` or `table = TRUE`, the previous list also contains the following element:

`PE.sd`	a named vector of standard errors of the estimated partial effects.

When both average and conditional partial effects are requested, two lists containing the previous elements are returned, indexed by the prefixes `ape` and `cpe`.

Author(s)

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References

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional response models", *Journal of Economic Surveys*, 25(1), 19-68.

Fang, K., & Ma, S. (2013), "Three-part model for fractional response variables with application to Chinese household health insurance coverage", *Journal of Applied Statistics*, 40(5), 925-940.

See Also

fracreg, for fitting fractional response models.
fracreg.reset and fracreg.ggoff, for specification tests.
fracreg.ptest, for non-nested hypothesis tests.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age,
           totemp = fracreg_k401k$totemp, sole = fracreg_k401k$sole)

m <- fracreg(y, X, type="1P", linkfrac="logit")
fracreg.pe(m)

### Simulated Examples

N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#Computing average partial effects for a logit fractional response model
res <- fracreg(y,X,linkfrac="logit",table=FALSE)
fracreg.pe(res)

#Computing average partial effects for a binary logit + fractional probit
#two-part model
res <- fracreg(y,X,linkbin="logit",linkfrac="probit",type="2P",inf=1,table=FALSE)
fracreg.pe(res)

#Computing conditional partial effects for X2 in the logit component
#of a two-part fractional response model, with the covariates evaluated
#at their median values
res <- fracreg(y,X,linkfrac="logit",type="2Pfrac",inf=1,table=FALSE)
fracreg.pe(res,APE=FALSE,CPE=TRUE,at="median",which.x="X2")

#Computing average partial effects for a three-part double-inflated model
```

```
y3p <- y
y3p[1:20] <- 0
y3p[21:40] <- 1
res3p <- fracreg(y3p,X,linkbin=c("logit","probit"),linkfrac="logit",type="3P",table=FALSE)
fracreg.pe(res3p)
```

`fracreg.ptest` *P Test for Fractional Response Models*

Description

`fracreg.ptest` is used to perform the P test to evaluate the specification of alternative, non-nested fractional response models by testing against each other.

Usage

```
fracreg.ptest(object1, object2, version = "Wald", table = TRUE)
```

Arguments

`object1`	an object containing the results of an `fracreg` command.
`object2`	an object containing the results of another `fracreg` command.
`version`	a vector containing the test versions to use. Available options: `Wald` (the default) and `LM`. Both options may be chosen at the same time and are computed in a robust way.
`table`	a logical value indicating whether a summary table with the test results should be printed.

Details

`fracreg.ptest` applies the P test statistic proposed by Davidson and MacKinnon (1981) to fractional response models estimated via`fracreg`. `fracreg.ptest` may be used to test against each other two alternative specifications for the link function in: (i) one-part fractional response models; (ii) the binary components of two-part and three-part fractional response models; (iii) the fractional components of two-part and three-part fractional response models; and (iv) two-part and three-part fractional response models.

P Test Framework: The P test allows the comparison of non-nested models (e.g., alternative link functions or non-nested regressors). Let model 1 specify $E_1(y|x) = G(x\beta)$ and model 2 specify $E_2(y|x) = H(z\theta)$. To test model 1 against model 2, the baseline model is augmented with the difference between the fitted values:

$$E(y|x) = G\left(x\beta + \gamma\left(\hat{y}_{M2} - \hat{y}_{M1}\right)\right)$$

where $\hat{y}_{M1} = G(x\hat{\beta})$ and $\hat{y}_{M2} = H(z\hat{\theta})$. The null hypothesis that model 1 is correct is tested via $H_0 : \gamma = 0$.

In addition, `fracreg.ptest` may be used to test one-part models against two-part or three-part models and in cases where the link functions are the same but the regressors are non-nested. See Ramalho, Ramalho and Murteira (2011) for details on the application of the P test in the fractional response framework.

Value

`fracreg.reset` returns a named vector with the test results.

Author(s)

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References

Davidson, R. and J.G. MacKinnon (1981), "Several tests for model specification on the presence of alternative hypotheses", *Econometrica*, 49(3), 781-793.

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional response models", *Journal of Economic Surveys*, 25(1), 19-68.

See Also

`fracreg`, for fitting fractional response models.
`fracreg.reset` and `fracreg.ggoff`, for specification tests.
`fracreg.pe`, for computing partial effects.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age,
           totemp = fracreg_k401k$totemp, sole = fracreg_k401k$sole)

m1 <- fracreg(y, X, type="1P", linkfrac="logit")
m2 <- fracreg(y, X, type="1P", linkfrac="probit")
fracreg.ptest(m1, m2)

### Simulated Examples

N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#Testing logit versus loglog specifications for standard fractional
#regression models using a LM version of the P test
res1 <- fracreg(y,X,linkfrac="logit",table=FALSE)
res2 <- fracreg(y,X,linkfrac="loglog",table=FALSE)
fracreg.ptest(res1,res2,"LM")

#Testing a logit one-part fractional response model versus a binary logit +
#fractional probit two-part model using a Wald version of the P test
res1 <- fracreg(y,X,linkfrac="logit",table=FALSE)
res2 <- fracreg(y,X,linkbin="logit",linkfrac="probit",type="2P",inf=1,table=FALSE)
fracreg.ptest(res1,res2,"Wald")
```

fracreg.reset *RESET Test for Fractional Response Models*

Description

`fracreg.reset` is used to perform the Regression Equation Specification Error Test (RESET) to check the functional form and specification of fractional response models.

Usage

```
fracreg.reset(object, lastpower.vec = 3, version = "LM", table = TRUE, ...)
```

Arguments

`object`	an object containing the results of an `fracreg` command.
`lastpower.vec`	a numeric vector containing the maximum powers of the linear predictors to be used in RESET tests.
`version`	a vector containing the test versions to use. Available options: `Wald`, `LM` (the default) and, only for the binary component of two-part models, `LR`. More than one option may be chosen.
`table`	a logical value indicating whether a summary table with the test results should be printed.
`...`	Arguments to pass to glm, which is used to estimate the model under the alternative hypothesis when `version` is a vector containing `"Wald"` or `"LR"`.

Details

`fracreg.reset` applies the RESET test statistic to fractional response models estimated via `fracreg`. `fracreg.reset` may be used to test the link specification of: (i) one-part fractional response models; (ii) the binary components of two-part and three-part fractional response models; and (iii) the fractional components of two-part and three-part fractional response models.

RESET Test Framework: The Regression Equation Specification Error Test (RESET) assesses whether the link function $G(\cdot)$ and the linear index $x\beta$ are correctly specified. It tests the null hypothesis $H_0 : \gamma = 0$ in the augmented model:

$$E(y|x) = G(x\beta + \sum_{k=2}^{P} \gamma_k (x\hat{\beta})^k)$$

where P is the maximum power of the linear predictor (specified by `lastpower.vec`) and $\hat{\beta}$ are the estimated parameters from the baseline model.

When the `Wald` version is implemented, it is taken into account the option that was chosen for computing standard errors in the model under evaluation. For the `LM` version, a robust version is computed in cases (i) and (iii) and a conventional version in case (ii). See Ramalho, Ramalho and Murteira (2011) for details on the application of the RESET test in the fractional response framework.

Value

`fracreg.reset` returns a named vector with the test results.

Author(s)

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References

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional response models", *Journal of Economic Surveys*, 25(1), 19-68.

Ramsey, J.B. (1969), "Tests for Specification Errors in Classical Linear Least-Squares Regression Analysis", *Journal of the Royal Statistical Society: Series B (Methodological)*, 31(2), 350-371.

See Also

fracreg, for fitting fractional response models.
fracreg.ggoff, for asymptotically equivalent specification tests.
fracreg.ptest, for non-nested hypothesis tests.
fracreg.pe, for computing partial effects.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age,
           totemp = fracreg_k401k$totemp, sole = fracreg_k401k$sole)

m <- fracreg(y, X, type="1P", linkfrac="logit")
fracreg.reset(m)

### Simulated Examples

N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#Testing the logit specification of a standard fractional response model
#using LM and Wald versions of the RESET test, based on 1 or 2 fitted powers of
#the linear predictor
res <- fracreg(y,X,linkfrac="logit",table=FALSE)
fracreg.reset(res,2:3,c("Wald","LM"))

#Testing the probit specification of the binary component of a two-part fractional
#regression model using LR-based RESET tests with quadratic and cubic fitted
#powers of the linear predictor
res <- fracreg(y,X,linkbin="probit",type="2Pbin",inf=1,table=FALSE)
fracreg.reset(res,3,"LR")
```

fracreghet *Fitting Fractional Response Models under Unobserved Heterogeneity*

Description

fracreghet is used to fit fractional response models under unobserved heterogeneity, i.e. regression models for proportions, percentages or fractions that suffer from neglected heterogeneity and/or endogeneity issues.

Usage

```
fracreghet(y, x, z = x, var.endog, start, type = "GMMx", link = "logit",
           intercept = T, table = T, variance = T,
           var.type = "robust", var.cluster, adjust = 0, ...)
```

Arguments

y	a numeric vector containing the values of the response variable.
x	a numeric matrix, with column names, containing the values of all covariates (exogenous and endogenous).
z	a numeric matrix, with column names, containing the values of all exogenous variables (covariates and instrumental variables). Defaults to x.
var.endog	a numeric vector containing the values of the endogenous covariate (or of some transformation of it), which will be used as dependent variable in the linear reduced form assumed for application of xv-type estimators.
start	a numeric vector containing the initial values for the parameters to be optimised. Optional.
type	a description of the estimator to compute: GMMx (the default), GMMxv, GMMz, LINx, LINxv, LINz or QMLxv.
link	a description of the link function to use. Available options for all estimators: logit and cloglog. Additional available options for QML and LIN estimators: probit, cauchit and loglog.
intercept	a logical value indicating whether the model should include a constant term or not.
table	a logical value indicating whether a summary table with the regression results should be printed.
variance	a logical value indicating whether the variance of the estimated parameters should be calculated. Defaults to TRUE whenever table = TRUE.
var.type	a description of the type of variance of the estimated parameters to be calculated. Options are robust, the default, and cluster.
var.cluster	a numeric vector containing the values of the variable that specifies to which cluster each observation belongs.
adjust	the numeric value to be added to the response variable in case of boundary observations when the LIN estimators are applied or the string drop, which implies that the boundary observations are dropped.
...	Arguments to pass to nlminb.

Details

`fracreghet` computes the GMM estimators proposed in Ramalho and Ramalho (2017) for fractional response models with unobserved heterogeneity: GMMx, which allows for neglected heterogeneity but not for endogeneity; GMMxv, which allows both issues and assumes a linear reduced form for the endogeneous covariate (or for a transformation of it); and GMMz, which also allows for both issues but does not require the assumption of a reduced form for the endogenous covariate. In addition, `fracreghet` also computes three linearised estimators (LINx, LINxv and LINz) that have similar features to their GMM counterparts. It also provides a QML estimator (QMLxv) that addresses endogeneity using a Control Function (CF) approach, which includes the first-stage reduced-form residuals as an additional regressor in the main fractional equation, providing a Hausman-type test for endogeneity.

Control Function (CF) Approach - QMLxv: When a continuous regressor y_{2i} is endogenous, the CF approach (Papke and Wooldridge, 2008; Terza et al., 2008) uses a two-stage procedure. First, a linear reduced form is estimated:

$$y_{2i} = z_i\pi + v_i$$

where z_i includes all exogenous variables and external instruments. The residuals $\hat{v}_i$ are then included in the fractional response model:

$$E(y_{1i}|z_i, y_{2i}, v_i) = G(x_i\beta + \gamma\hat{v}_i)$$

A test of $H_0 : \gamma = 0$ serves as a robust Hausman-type test for endogeneity.

Generalised Method of Moments (GMM): For estimators like GMMz, which do not strictly require a linear reduced form, the estimation relies on population orthogonality conditions between the instruments Z_i and the model residuals:

$$E[Z_i(y_i - G(x_i\beta))] = 0$$

or via specific transformations of the dependent variable to eliminate unobserved heterogeneity (Ramalho and Ramalho, 2017).

For overidentified models, `fracreghet` calculates Hansen's J statistic. For `GMMx` and `LINx`, `fracreghet` stores the information needed to implement the RESET test (fracreghet.reset). For all estimators, `fracreghet` stores the information needed to calculate partial effects (fracreghet.pe).

Value

`fracreghet` returns a list with the following elements:

`class`	"fracreghet".
`formula`	the model formula.
`type`	the name of the estimator computed.
`link`	the name of the specified link.
`adjust`	The value or the type of the adjustment applied to LIN estimators.
`p`	a named vector of coefficients.
`Hy`	the transformed values of the response variable when GMM or LIN estimators are computed or the values of the response variable in the QML case.
`xbhat`	the fitted mean values of the linear predictor (for xv-type estimators, includes the term relative to the first-stage residual).
`converged`	logical. Was the algorithm judged to have converged?
`x.names`	a vector containing the names of the covariates.

In case of an overidentifying model, the following element is also returned:

`J` the result of Hansen's J test of overidentifying moment conditions.

If `variance = TRUE` or `table = TRUE` and the algorithm converged successfully, the previous list also contains the following elements:

`p.var` a named covariance matrix.

`var.type` covariance matrix type.

If `var.type = "cluster"`, the list also contains the following element:

`var.cluster` the variable that specifies to which cluster each observation belongs.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Papke, L. E. and Wooldridge, J. M. (2008), "Panel data methods for fractional response variables with an application to test pass rates", *Journal of Econometrics*, 145, 121-133.

Ramalho, E. A., & Ramalho, J. J. S. (2017), "Moment-based estimation of nonlinear regression models with boundary outcomes and endogeneity, with applications to nonnegative and fractional responses", *Econometric Reviews*, 36(4), 397-420.

Terza, J. V., Basu, A., and Rathouz, P. J. (2008), "Two-stage residual inclusion estimation: addressing endogeneity in health econometric modeling", *Journal of Health Economics*, 27(3), 531-543.

See Also

`fracreghet.reset`, for the RESET test.
`fracreghet.pe`, for computing partial effects.
`fracreg`, for fitting standard cross-sectional fractional response models.
`fracregpd`, for fitting panel data fractional response models.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X_het <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age)

# fracreghet estimators do not allow exact 1s or 0s
y_adj <- y
y_adj[y_adj == 1] <- 0.999

# Artificial instrumental variable for demonstration
set.seed(42)
Z_emp <- cbind(X_het, z = fracreg_k401k$mrate * rnorm(length(y)))
fracreghet(y_adj, X_het, Z_emp, var.endog = X_het[, "mrate"], type="QMLxv", link="logit")

### Simulated Examples

set.seed(123)
N <- 1000
```

```
x1 <- rnorm(N)

# Simulating an endogenous variable (var.endog) and an instrument (z1)
z1 <- rnorm(N)
u <- 0.5 * z1 + rnorm(N)
var.endog <- 0.8 * z1 + u
y_endog <- exp(0.5 * x1 + 1.2 * var.endog + u) / (1 + exp(0.5 * x1 + 1.2 * var.endog + u))

# Avoid exact 0 or 1 boundaries for some estimators
y_endog[y_endog <= 0] <- 0.01
y_endog[y_endog >= 1] <- 0.99

X <- cbind(x1 = x1, var.endog = var.endog)
Z <- cbind(x1 = x1, z1 = z1)

# Exogeneity (assuming var.endog is exogenous for comparison), GMMx estimator
fracreghet(y = y_endog, x = X, type = "GMMx", link = "logit")

# Endogeneity, GMMz estimator (does not require reduced form for endog)
fracreghet(y = y_endog, x = X, z = Z, type = "GMMz", link = "logit")

# Endogeneity, GMMxv estimator (assumes linear reduced form for var.endog)
fracreghet(y = y_endog, x = X, z = Z, var.endog = var.endog, type = "GMMxv", link = "logit")

# Endogeneity, QMLxv control function approach
fracreghet(y = y_endog, x = X, z = Z, var.endog = var.endog, type = "QMLxv", link = "logit")
```

`fracreghet.pe` *Fractional Response Models under Unobserved Heterogeneity - Partial Effects*

Description

`fracreghet.pe` is used to compute average and/or conditional partial effects in fractional response models under unobserved heterogeneity.

Usage

```
fracreghet.pe(object, smearing = T, APE = T, CPE = F, at = NULL,
              which.x = NULL, table = T, variance = T)
```

Arguments

`object`	an object containing the results of an `fracreghet` command.
`smearing`	a logical value indicating whether the smearing correction is to be applied
`APE`	a logical value indicating whether average partial effects are to be computed.
`CPE`	a logical value indicating whether conditional partial effects are to be computed.
`at`	a numeric vector containing the covariates' values at which the conditional partial effects are to be computed or the strings `"mean"` (the default) or `"median"`, in which cases the covariates are evaluated at their mean or median values (or mode, in case of dummy variables), respectively.

`which.x`	a vector containing the names of the covariates to which the partial effects are to be computed.
`table`	a logical value indicating whether a summary table with the results should be printed.
`variance`	a logical value indicating whether the variance of the estimated partial effects should be calculated. Defaults to `TRUE` whenever `table = TRUE`.

Details

`fracreghet.pe` calculates partial effects for fractional response models estimated via `fracreghet`. `fracreghet.pe` may be used to compute average or conditional partial effects. These partial effects may be conditional only on observables, using the smearing estimator, or also on unobservables, setting the error term to zero.

Partial Effects under Unobserved Heterogeneity: When unobserved heterogeneity or endogeneity is present, calculating partial effects requires dealing with the unobserved error v_i. Let the conditional mean be $E(y|x, v) = G(x\beta + \gamma v)$. - **Conditional on Observables (Smearing):** The unobserved heterogeneity is integrated out over its empirical distribution. The average partial effect for a continuous variable x_k is computed as:

$$PE_k(x) = \frac{1}{N}\sum_{i=1}^{N} g(x\beta + \gamma\hat{v}_i)\beta_k$$

- **Conditional on Unobservables (Error = 0):** The partial effect is evaluated for an individual with the mean level of unobserved heterogeneity ($v = 0$):

$$PE_k(x) = g(x\beta)\beta_k$$

For discrete variables, the partial effects are calculated as the discrete differences evaluated using either the smearing approach or setting the error term to zero.

For calculating standard errors, it is taken into account the option that was previously chosen for estimating the model. See Ramalho and Ramalho (2017) for details on the computation of partial effects for fractional response models under unobserved heterogeneity.

Value

`fracreghet.pe` returns a list with the following element:

`PE.p`	a named vector of partial effects.

If `variance = TRUE` or `table = TRUE`, the previous list also contains the following element:

`PE.sd`	a named vector of standard errors of the estimated partial effects.

When both average and conditional partial effects are requested, two lists containing the previous elements are returned, indexed by the prefixes `ape` and `cpe`.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Ramalho, E. A., & Ramalho, J. J. S. (2017), "Moment-based estimation of nonlinear regression models with boundary outcomes and endogeneity, with applications to nonnegative and fractional responses", *Econometric Reviews*, 36(4), 397-420.

See Also

fracreghet, for fitting fractional response models under unobserved heterogeneity.
fracreghet.reset, for the RESET test.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X_het <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age)

# fracreghet estimators do not allow exact 1s or 0s
y_adj <- y
y_adj[y_adj == 1] <- 0.999

# Artificial instrumental variable for demonstration
set.seed(42)
Z_emp <- cbind(X_het, z = fracreg_k401k$mrate * rnorm(length(y)))
res_emp <- fracreghet(y_adj, X_het, Z_emp, var.endog = X_het[, "mrate"],
                      type="QMLxv", link="logit", table=FALSE)
fracreghet.pe(res_emp, which.x="mrate")

### Simulated Examples

N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

Z <- cbind(rnorm(N),rnorm(N),rnorm(N))
dimnames(Z)[[2]] <- c("Z1","Z2","Z3")

y <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))

res <- fracreghet(y,X,type="GMMx",table=FALSE)

#Smearing estimator of average partial effects for variable X1
fracreghet.pe(res,which.x="X1")

#Naive estimator of conditional partial effects for all covariates,
#which are evaluated at X1=1 and X2=-1
fracreghet.pe(res,smearing=FALSE,APE=FALSE,CPE=TRUE,at=c(1,-1))
```

---

fracreghet.reset *RESET Test for Fractional Response Models under Neglected Heterogeneity*

Description

fracreghet.reset is used to test the specification of fractional response models estimated by GMMx or LINx.

Usage

```
fracreghet.reset(object, lastpower.vec = 3, version = "Wald", table = T, ...)
```

Arguments

`object`	an object containing the results of an `fracreghet` command.
`lastpower.vec`	a numeric vector containing the maximum powers of the linear predictors to be used in RESET tests.
`version`	a vector containing the test versions to use. Available options: `Wald` (the default) and `LM` (only available for `GMMx`).
`table`	a logical value indicating whether a summary table with the test results should be printed.
`...`	Arguments to pass to nlminb, which is used to estimate the model under the alternative hypothesis when `version` is equal to `"Wald"` and the null model was estimated by `GMMx`.

Details

`fracreghet.reset` applies the RESET test statistic to fractional response models estimated via `fracreghet` using the options `GMMx` or `LINx`. `fracreghet.reset` may be used to test simultaneously the validity of the link specification and the transformation applied to the response variable by each estimator.

RESET Test under Unobserved Heterogeneity: The test is based on augmenting the original model with powers of the linear predictor $x\hat{\beta}$. For GMMx, it tests $H_0 : \gamma = 0$ in the expanded moment conditions:

$$E\left[Z_i\left(H(y_i) - \exp\left(x_i\beta + \sum_{k=2}^{P}\gamma_k(x_i\hat{\beta})^k\right)E(e^{c_i})\right)\right] = 0$$

This simultaneously evaluates whether the mean function and the specific heterogeneity transformation $H(\cdot)$ are correctly specified.

It is taken into account the option that was chosen for computing standard errors in the model under evaluation. See Ramalho and Ramalho (2017) for details.

Value

`fracreghet.reset` returns a named vector with the test results.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Ramalho, E. A., & Ramalho, J. J. S. (2017), "Moment-based estimation of nonlinear regression models with boundary outcomes and endogeneity, with applications to nonnegative and fractional responses", *Econometric Reviews*, 36(4), 397-420.

Ramsey, J.B. (1969), "Tests for Specification Errors in Classical Linear Least-Squares Regression Analysis", *Journal of the Royal Statistical Society: Series B (Methodological)*, 31(2), 350-371.

See Also

fracreghet, for fitting fractional response models under unobserved heterogeneity.
fracreghet.pe, for computing partial effects.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X_het <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age)

# fracreghet estimators do not allow exact 1s or 0s
y_adj <- y
y_adj[y_adj == 1] <- 0.999

# Artificial instrumental variable for demonstration
set.seed(42)
Z_emp <- cbind(X_het, z = fracreg_k401k$mrate * rnorm(length(y)))
res_emp <- fracreghet(y_adj, X_het, type="GMMx", link="logit", table=FALSE)
fracreghet.reset(res_emp)

### Simulated Examples

N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

Z <- cbind(rnorm(N),rnorm(N),rnorm(N))
dimnames(Z)[[2]] <- c("Z1","Z2","Z3")

y <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))

res <- fracreghet(y,X,type="GMMx",table=FALSE)

#LM and Wald versions of the RESET test, based on 1 or 2 fitted powers of xb
fracreghet.reset(res,2:3,c("Wald","LM"))
```

---

fracregpd *Fitting Panel Data Fractional Response Models*

Description

fracregpd is used to fit panel data regression models when the dependent variable has a bounded, fractional nature.

Usage

```
fracregpd(id, time, y, x, z, var.endog, x.exogenous = T, lags, start, type,
 GMMww.cor = T, link = "logit", intercept = T, table = T, variance = T,
 var.type = "cluster", tdummies = F, bootstrap = F, B = 200, ...)
```

Arguments

`id`	a numeric vector identifying the cross-sectional units.
`time`	a numeric vector identifying the time periods in which the cross-sectional units were observed.
`y`	a numeric vector containing the values of the response variable.
`x`	a numeric matrix, with column names, containing the values of all covariates (exogenous and endogenous).
`z`	a numeric matrix, with column names, containing the values of all exogenous variables (covariates and external instrumental variables). Only required in case of endogenous explanatory variables.
`var.endog`	a numeric vector containing the values of the endogenous covariate (or of some transformation of it), which will be used as dependent variable in the linear reduced form assumed for application of the `QMLcre` estimator. Only required for this estimator.
`x.exogenous`	a logical value indicating whether all explanatory variables are assumed to be exogenous or not.
`lags`	a logical value indicating whether the first lags of `x` or `z` should be used as instruments for `x`. Defaults to `TRUE` for the GMMww and GMMc estimators and to `FALSE` for the remaining estimators. The `GMMcre` and `QMLcre` estimators do not admit lagged instruments.
`start`	a numeric vector containing the initial values for the parameters to be optimised. Optional.
`type`	a description of the estimator to compute: `GMMww`, `GMMc`, `GMMbgw`, `GMMpfe`, `GMMcre`, `GMMpre` or `QMLcre`.
`GMMww.cor`	a logical value indicating whether each explanatory variable should be transformed in deviations from its overall mean before computing the `GMMww` estimator.
`link`	a description of the link function to use. Available options for all GMM estimators: `logit` and `cloglog`. Only option for the `QMLcre` estimator: `probit`.
`intercept`	a logical value indicating whether the model should include a constant term or not. Only relevant for the `GMMpre` estimator.
`table`	a logical value indicating whether a summary table with the regression results should be printed.
`variance`	a logical value indicating whether the variance of the estimated parameters should be calculated. Defaults to `TRUE` whenever `table = TRUE`.
`var.type`	a description of the type of variance of the estimated parameters to be calculated. Options are `cluster`, the default, and `robust`. In overidentified models, it also affects the parameter estimates via the GMM weighting matrix.
`tdummies`	a logical value indicating whether time dummies should be included among the model explanatory variables.
`bootstrap`	a logical value indicating whether bootstrap should be used in the estimation of the parameter standard errors.
`B`	the number of bootstrap replications.
`...`	Arguments to pass to nlminb.

Details

`fracregpd` computes the GMM estimators proposed in Ramalho, Ramalho and Coelho (2018) for panel data fractional response models with both time-variant and time-invariant unobserved heterogeneity and endogeneous covariates: GMMww, GMMc, GMMbgw, GMMpfe, GMMcre and GMMpre. In addition, `fracregpd` also computes QMLcre, which was proposed by Papke and Wooldridge (2008) and Wooldridge (2019).

Correlated Random Effects (CRE) - QMLcre: In panel data, unobserved individual-specific heterogeneity c_i may be correlated with the covariates x_{it}. The CRE approach (Papke and Wooldridge, 2008) models this dependence by projecting c_i onto the time averages of the strictly exogenous covariates $\bar{x}_i$:

$$c_i = \psi + \bar{x}_i \xi + a_i$$

where a_i is an error term independent of x_i. Assuming $a_i | x_i \sim N(0, \sigma_a^2)$ and a probit link, integrating out a_i yields the "population-averaged" or scaled conditional mean:

$$E(y_{it}|x_i) = G(x_{it}\beta_a + \psi_a + \bar{x}_i \xi_a)$$

where the parameters with subscript a are scaled by $(1 + \sigma_a^2)^{-1/2}$. This equation is estimated via pooled Bernoulli QML.

Generalised Method of Moments (GMM): For models where strict exogeneity fails or the link function is an exponential-type link, Ramalho et al. (2018) propose GMM estimators based on the following general moment conditions:

$$E[Z_{it}(H(y_{it}) - \exp(x_{it}\beta + c_i))] = 0$$

where $H(\cdot)$ is a transformation function and Z_{it} is a matrix of valid instruments. Estimators such as GMMww, GMMc, and GMMbgw use different transformations to eliminate the unobserved fixed effect c_i before applying GMM.

For overidentified models, `fracregpd` calculates Hansen's J statistic to test the validity of the overidentifying restrictions.

Value

`fracregpd` returns a list with the following elements:

`type`	the name of the estimator computed.
`link`	the name of the specified link.
`p`	a named vector of coefficients.
`Hy`	the transformed values of the response variable when GMM estimators are computed or the values of the response variable in the QML case.
`converged`	logical. Was the algorithm judged to have converged?

In case of an overidentifying model, the following element is also returned:

`J`	the result of Hansen's J test of overidentifying moment conditions.

If `variance = TRUE` or `table = TRUE` and the algorithm converged successfully, the previous list also contains the following elements:

`p.var`	a named covariance matrix.
`var.type`	covariance matrix type.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Papke, L. and Wooldridge, J.M. (2008), "Panel data methods for fractional response variables with an application to test pass rates", *Journal of Econometrics*, 145(1-2), 121-133.

Ramalho, E. A., Ramalho, J. J. S., & Coelho, L. M. S. (2018), "Exponential Regression of Fractional-Response Fixed-Effects Models with an Application to Firm Capital Structure", *Journal of Econometric Methods*, 7(1), 20150019.

Wooldridge, J. M. (2019). Correlated random effects models with unbalanced panels. *Journal of Econometrics*, 211(1), 137-150.

See Also

fracreg, for fitting standard cross-sectional fractional response models.
fracreghet, for fitting cross-sectional fractional response models with unobserved heterogeneity.

Examples

```
### Empirical 401(k) Examples
data("fracreg_k401k")
y <- fracreg_k401k$prate
X <- cbind(mrate = fracreg_k401k$mrate, age = fracreg_k401k$age,
           totemp = fracreg_k401k$totemp, sole = fracreg_k401k$sole)

# Artificial panel data structure for demonstration
N_emp <- nrow(X)
id_emp <- rep(1:(N_emp/2), each=2)
time_emp <- rep(1:2, times=N_emp/2)
fracregpd(id_emp, time_emp, y, X, type="QMLcre", link="probit")

### Simulated Examples

set.seed(123)
# Simulating Panel Data
N <- 100
T_periods <- 5
id <- rep(1:N, each = T_periods)
time <- rep(1:T_periods, times = N)
x_panel <- rnorm(N * T_periods)

# Unobserved individual effect (CRE)
c_i <- rep(rnorm(N), each = T_periods)
y_panel <- exp(x_panel + c_i) / (1 + exp(x_panel + c_i))

X <- cbind(x_panel = x_panel)

# Endogenous variable and instrument simulation
z_panel <- rnorm(N * T_periods)
u_panel <- 0.5 * z_panel + rnorm(N * T_periods)
var_endog <- 0.8 * z_panel + u_panel
y_endog <- exp(x_panel + 1.2 * var_endog + c_i + u_panel) /
             (1 + exp(x_panel + 1.2 * var_endog + c_i + u_panel))
```

```
X_endog <- cbind(x_panel = x_panel, var_endog = var_endog)
Z_inst <- cbind(x_panel = x_panel, z_panel = z_panel)

# Estimate a Correlated Random Effects (CRE) Model
fracregpd(id=id, time=time, y=y_panel, x=X, type="QMLcre", link="probit")

# Exogeneity, no lags, no time dummies, clustered standard errors, GMMbgw estimator
fracregpd(id=id, time=time, y=y_panel, x=X, type="GMMbgw")

# Lagged covariates and instruments, robust standard errors, GMMww estimator
fracregpd(id=id, time=time, y=y_panel, x=X, lags=TRUE, type="GMMww", var.type="robust")

# Endogeneity, time dummies, GMMpfe estimator
fracregpd(id=id, time=time, y=y_endog, x=X_endog, z=Z_inst,
          x.exogenous=FALSE, type="GMMpfe", tdummies=TRUE)
```

`fracreg_k401k` *401(k) Plan Participation Data*

Description

A cross-sectional dataset on 401(k) plan participation rates and firm characteristics, widely used in empirical applications of fractional response models (e.g., Papke & Wooldridge, 1996). The data is derived from the 'wooldridge' package.

Usage

```
data("fracreg_k401k")
```

Format

A data frame with 1,534 observations and 10 variables:

prate Participation rate: proportion of eligible employees participating in the 401(k) plan (0 to 1).

mrate Match rate: the firm's contribution matching rate per dollar.

totpart Total number of participants.

totelg Total number of eligible employees.

age Age of the 401(k) plan in years.

totemp Total number of firm employees.

sole Indicator variable: 1 if the 401(k) is the sole retirement plan offered.

ltotemp Natural log of total employees.

age_sq Square of plan age.

mrate_sq Square of match rate.

Source

Papke, L. E., & Wooldridge, J. M. (1996). "Econometric Methods for Fractional Response Variables with an Application to 401(k) Plan Participation Rates." Journal of Applied Econometrics, 11(6), 619-632.

Examples

```
data("fracreg_k401k")
summary(fracreg_k401k$prate)
```

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